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📅 1999 Oct 19

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🔄 Personal Project

♀️ Female - She/Her

👤 Profile

I am currently a PhD student at Sapienza, University of Rome, M.Sc. graduated in Computer Engineering, specialized in Artificial Intelligence and Intelligent Robotics, with huge passion in the field of AI, in particular towards artificial vision and generative AI. Thanks to my involvement in various educational and external projects, I have had the opportunity to expand my technical knowledge and develop skills in teamwork, as well as in the analysis and resolution of problems.

📁 Work Experience

2023 Mar – 2023 Aug
Salerno, Italia

University Research Scholarship.

RFI & Università degli Studi di Salerno

Semester research scholarship in collaboration between the University of Salerno and R.F.I. for the design and implementation of artificial vision algorithms for autonomous driving in the railway field.

Design of the on board hardware acquisition system.

Design, analysis and validation of artificial vision algorithms for real-time anomaly detection along the railway track.

🎓 Education

2025 Nov – Present
Austin, TX, USA

Visiting Scholar

UT Austin

Visiting period under the supervision of Prof. Adam Klivans. The research applies Representation Learning and Mechanistic Interpretability principles to protein engineering.

2023 Nov – present
Rome, Italy

PhD student in the National PhD AI for Government and public security

La Sapienza University of Rome

The research realm involves developing practical defense techniques to overcome the problem of media manipulation, caused by the widespread of generative AI.

2021 Sep – 2023 Sep
Fisciano, Italia

Master's Degree (LM-32) in Computer Engineering and Electrical Engineering and Applied Mathematics.

Università degli Studi di Salerno

Master's Degree in topics of Artificial Intelligence and Intelligent Robotics, currently in progress. The expected graduation date is September 2023, with a thesis titled "Automatic Generation of Identity-Preserving Gesture Videos", with minimum expected grade 110.

Below is the updated list of completed exams with their respective grades.

Weighted average grade: 29.42

- Cognitive Robotics: 30L
- Algorithms and Protocols for Security: 30
- Decision Support Systems: 30
- Automation: 25
- Design and Analysis of Algorithms: 30
- High Performance Computing: 30L
- Machine Learning: 30
- Embedded Systems: 28
- Semantic Technologies: 30
- Software Engineering: 30
- Artificial Vision: 30L
- Autonomous Vehicle Driving: 30

2018 Sep – 2021 Jul
Fisciano, Italia

Bachelor's Degree (L-8) in Computer Engineering and Electrical Engineering and Applied Mathematics.

Università degli Studi di Salerno

Bachelor's Degree (L-8) in Computer Engineering and Electrical Engineering and Applied Mathematics, completed with a final grade of 110/110 cum laude. The thesis titled "Automatic Speaker Recognition Using a Neural Network Integrated into a Web Portal" was submitted.

Below is the updated list of completed exams with their respective grades.

Weighted average grade: 29.23

- Mathematical Analysis 1: 30
- Physics 1: 30L
- Physics 2: 30L
- Fundamentals of Programming: 29

- Geometry, Algebra, and Logic: 30L
- Programming Laboratory: 30
- Logic Networks: 30L
- Algorithms and Data Structures: 27
- Mathematical Analysis 2: 28
- Computer Architecture and Operating Systems: 29
- Electrical Engineering: 30
- Fundamentals of Automatic Controls: 30L
- Digital Technologies: 30
- Signal Theory: 30L
- Antennas and Wireless Communications: 30
- Databases: 28
- Object-Oriented Programming: 29
- Computer Networks: 30
- Computer Control System Technologies: 23
- Web Software Technologies: 30
- Telecommunications Theory and Techniques: 30L

2013 Sep – 2018 Jul
Angri, Italia

Scientific High School Diploma.

Liceo Don Carlo La Mura

The study program was completed with a final grade of 100/100 cum laude.

Skills

- Proactive.
- Problem Solving.
- Team Working.
- Tenacity in pursuing a goal.
- Easy to resolve conflicts.
- Highly motivated.

Languages

French
Beginner

English

Certifications issued by Cambridge Assessment.
B2 level - First in 2018
C1 level - Advanced in 2021.

Awards

2023 May

Participation in courses reserved for excellent students.

Università degli Studi di Salerno

Participation exclusively reserved for students with an exam grade average of at least 28 with at maximum a single left exam. The courses focused on the following topics:

- Research and Innovation: Methods, Tools, and Professional Figures
- Startup Creation: Challenges and Opportunities
- Ethical AI

2023 Mar

Recognition for an outstanding academic career.

Università degli Studi di Salerno

For achieving a Bachelor's Degree in Computer Engineering with the highest honors and in the shortest possible time at the University of Salerno.

Technical Skills

- Python language.
- C language.
- R language.
- Tensorflow.
- Rasa conversational platform.
- Agile Methodology.
- Java language.
- Matlab.
- ROS noetic framework.
- Pytorch.
- Django framework.
- Linux.

Publications

2026

CIRCE: Chemistry-Informed Reaction-Catalyst Embeddings for Enzyme Retrieval

Under Submission

Enzymes can make chemical production more selective and sustainable, but identifying enzymes for desired reactions and characterizing enzyme promiscuity remain major bottlenecks.

Framing this task as reaction-enzyme retrieval is especially challenging because the useful sequence signal may be localized to a small, often unannotated functional region, while reaction annotations may be incomplete and enzyme promiscuity can create multiple valid matches.

We introduce CIRCE (Chemistry-Informed Reaction–Catalyst Embeddings), a single-stage dual encoder that represents these two modalities with functional-site-guided protein pooling and molecular-set reaction chemistry.

On the official ReactZyme benchmark, CIRCE reaches a macro-averaged mean reciprocal rank of 0.6489 across six settings, compared with 0.5735 for TIGER ESM2Text.

These results show that chemistry- and site-informed representations, through chemically and biologically motivated inductive biases, can improve enzyme-reaction retrieval.

2026

Architectural Backdoors in Vision-Language Model Supply Chains via Representation Steering

Under Submission

Vision–Language Models (VLMs) are increasingly deployed through a model supply chain in which pretrained checkpoints, architecture definitions, text encoders, and ex-ported computation graphs are distributed by third parties and reused across downstream services. This reuse model creates a security-critical trust boundary: VLM deployments inherit not only learned parameters but also executable behavior encoded in shared model artifacts. In this paper, we show that a malicious provider can exploit this trust boundary by embedding architectural backdoors into VLM supply chains through representation steering. Our attack introduces dormant steering logic into the model architecture through a trigger-gated additive modification of an intermediate representation, without poisoning training data, controlling downstream fine-tuning, or modifying prompts at deployment time. When the trigger is absent, the modification reduces to zero and the model follows its normal computation, preserving clean utility. When the trigger is present, a steering direction shifts the internal representation toward an attacker-defined objective.

We evaluate the attack across multiple VLM families, including CLIP, BLIP, T5, LLaVA, Qwen3-VL, and diffusion-based architectures. We evaluate the attack across multiple downstream tasks, including visual question answering, text-to-image generation, retrieval, and semantic response biasing, showing that the same architectural steering primitive compromises integrity, safety enforcement, and ranking fairness while preserving normal behavior on clean inputs. These results show that shared VLM artifacts can carry dormant steering logic that acts as an architectural backdoor against downstream services. Representation steering is a practical architectural backdoor payload for VLM supply chains, and securing VLM deployments requires auditing the executable logic distributed with model artifacts, rather than only their weights.

2026

Anchored Protein Engineering

Under Submission

Inspired by recent work on discrete diffusion for anchored unmasking, we introduce a new training-free framework for protein engineering that can generate higher-order variants that improve one or more phenotypes while exhibiting favorable epistatic interactions. Prior approaches require mutation sites to be specified in advance and can drive sequences away from the natural protein manifold. We address these limitations with Anchored Protein Engineering via quantized eXpectation (APEX), a decoding-time method consisting of anchor selection, tilted decoding, and iterative refinement. APEX selects anchor residues using a reward-gradient score with an explicit correction for pairwise epistasis, decodes substitutions for anchors by tilting the base model’s per-site predictive distributions along reward gradients, and revisits low-scoring anchors iteratively for optional test-time scaling. All phases query the base model only through masked-conditional logits, a primitive shared by masked language models, discrete-diffusion models, and inverse-folding networks; as a result, APEX works out of the box across commonly used model families. Across various protein-engineering benchmarks, including thermostability and solubility, APEX achieves state-of-the-art results and avoids the predictor-gaming artifacts that reward-search-only baselines tend to produce.

2026

Mind the Modality Gap: From Constitutional AI to Contextual Safety Through Language Bridging

Under Submission

Safety is context-dependent across cultures, jurisdictions, and use cases, yet existing alignment approaches embed a fixed notion of safety and require a separate bespoke pipeline per modality. We propose AnyCAI, a framework that generalizes Constitutional AI’s generate-critique-revise paradigm to any generative modality through a language bridge: safety principles are expressed as natural-language constitutions, a language-capable model critiques outputs regardless of their modality, and a language-only reviser rewrites the prompt accordingly. We validate AnyCAI across four modalities. On text-to-image, text-to-video, and vision-language generation, AnyCAI Pareto-dominates every safety-aligned baseline: (1) text-to-image generation, where HR improves from 42.6% to 90.3% while preserving image fidelity; (2) text-to-video generation, where HR improves from 75.6% to 96.3% with minimal utility penalty; (3) vision-language model, where HR improves from 85.1% to 99.0%; and (4) text-to-music generation, where AnyCAI demonstrates proof-of-concept alignment, improving HR from 0% to 20% with near-zero false refusal. To properly measure contextual definitions of safety, we further propose a constitution-driven evaluation protocol that auto-generates standardised test sets from any constitution, eliminating the need for human-curated datasets that cannot keep pace with policy changes.

2026

Neutralizing Proactive Defense using Diffusion-based Upsampling*Proceedings of the 2026 ACM Workshop on Information Hiding and Multimedia Security*

The rapid spread of open-source generative models has made it easy to create highly realistic manipulated media, posing a critical threat to content authenticity and provenance. Proactive image protection mechanisms have recently emerged as a promising defense, embedding imperceptible or characteristic signals into images to enable reliable manipulation detection. However, their robustness under realistic and adversarial post-release conditions remains largely unexplored. In this work, we present a systematic evaluation of the robustness of recent state-of-the-art proactive image protection schemes in a black-box setting. We analyze the resilience of PADL and DiffVax protections against a broad range of attacks, including classical image transformations and diffusion-based reconstruction attacks that implicitly re-synthesize image content while preserving perceptual quality. Our results reveal that, despite strong performance under limited perturbations, current proactive defenses are vulnerable to unseen image manipulations and generative reconstruction attacks. Considering PADL, we empirically demonstrate that adding diffusion-based upsampling attacks in the training does not improve robustness, without increasing protection intensity. These findings expose critical gaps between assumed and real-world threat models, highlighting the need for more robust proactive protection designs and standardized evaluation protocols for trustworthy digital media.

2026

Folding scFv-Antigen Complexes at Scale*GEM 2026 - ICLR Workshop*

Accurate modeling of antibody-antigen (Ab-Ag) complexes is central to biologic development, yet the reliability and failures of modern Ab-Ag folding pipelines remain poorly characterized. Single-chain variable fragments (scFvs) are therapeutically important antibodies, but large-scale evaluations of structure prediction models on scFv-Ag complexes are largely lacking. We introduce a scalable benchmarking pipeline that generates large ensembles of scFv-Ag structure predictions by cofolding a curated subset of 3,800 Ab-Ag complexes from SAbDab using multiple state-of-the-art models under diverse inference-time settings. The resulting dataset, **SCALE scFv-Ag CompLex** includes standardized scFv-Ag sequences and around 200,000 predicted complexes spanning different models, sampling strategies, and auxiliary inputs. Using **SCALE**, we evaluate model performance in recovering correct scFv-Ag interfaces and assess the ability of existing confidence metrics to select the best structure from prediction ensembles. We find that while confidence scores effectively distinguish easy from hard scFv-Ag complexes, they often fail to identify the highest-quality interface for a given target. Further analysis shows that correct binding modes are often present but rare, and inference-time choices like sampling, recycling, and using evolutionary or structural information are crucial for accurate scFv-Ag complex predictions.

2026

Semantic Steering via Hyperbolic Geometry*ECCV 2026 Beyond Euclidean Workshop- <https://sites.google.com/view/beyondeuclidean>; under submission*

As modern text-to-image (T2I) models draw closer to synthesizing highly realistic content, the threat of unsafe content generation grows, and it becomes paramount to exercise control. Existing approaches steer these models by applying Euclidean adjustments to text embeddings, redirecting the generation away from unsafe concepts. In this work, we introduce hyperbolic control (HyCon): a novel control mechanism based on parallel transport that leverages semantically aligned hyperbolic representation space to yield more expressive and stable manipulation of concepts. HyCon reuses off-the-shelf generative models and a state-of-the-art hyperbolic text encoder, linked via a lightweight adapter. HyCon achieves state-of-the-art results across four safety benchmarks and four T2I backbones, showing that hyperbolic steering is a practical and flexible approach for more reliable T2I generation.

2026

Harnessing Hyperbolic Geometry for Harmful Prompt Detection and Sanitization*ICLR2026*

Vision-Language Models (VLMs) have become essential for tasks such as image synthesis, captioning, and retrieval by aligning textual and visual information in a shared embedding space. Yet, this flexibility also makes them vulnerable to malicious prompts designed to produce unsafe content, raising critical safety concerns. Existing defenses either rely on blacklist filters, which are easily circumvented, or on heavy classifier-based systems, both of which are costly and fragile under embedding-level attacks. We address these challenges with two complementary components: Hyperbolic Prompt Espial (HyPE) and Hyperbolic Prompt Sanitization (HyPS). HyPE is a lightweight anomaly detector that leverages the structured geometry of hyperbolic space to model benign prompts and detect harmful ones as outliers. HyPS builds on this detection by applying explainable attribution methods to identify and selectively modify harmful words, neutralizing unsafe intent while preserving the original semantics of user prompts. Through extensive experiments across multiple datasets and adversarial scenarios, we prove that our framework consistently outperforms prior defenses in both detection accuracy and robustness. Together, HyPE and HyPS offer an efficient, interpretable, and resilient approach to safeguarding VLMs against malicious prompt misuse.

2025

What is Adversarial Training for Diffusion Models?*ICLR2026*

We answer the question in the title, showing that adversarial training (AT) for diffusion models (DMs) fundamentally differs from classifiers: while AT in classifiers enforces output invariance, AT in DMs requires equivariance to keep the diffusion process aligned with the data distribution. AT is a way to enforce smoothness in the diffusion flow, improving robustness to outliers and corrupted data. Unlike prior art, our method makes no assumptions about the noise model and integrates seamlessly into diffusion training by adding random noise, similar to randomized smoothing, or adversarial noise, akin to AT. This enables intrinsic capabilities such as handling noisy data, dealing with extreme variability such as outliers, preventing memorization, and improving robustness. We rigorously evaluate our approach with proof-of-concept datasets with known distributions in low- and high-dimensional space, thereby taking a perfect measure of errors; we further evaluate on standard benchmarks such as CIFAR-10, CelebA and LSUN Bedroom, showing strong performance under severe noise, data corruption, and iterative adversarial attacks.

2025

Implicit Inversion turns CLIP into a Decoder*ICLR2026*

CLIP is a discriminative model trained to align images and text in a shared embedding space. Due to its multimodal structure, it serves as the backbone of many generative pipelines, where a decoder is trained to map from the shared space back to images. In this work, we show that image synthesis is nevertheless possible using CLIP alone -- without any decoder, training, or fine-tuning. Our approach optimizes a frequency-aware implicit neural representation that encourages coarse-to-fine generation by stratifying frequencies across network layers. To stabilize this inverse mapping, we introduce adversarially robust initialization, a lightweight Orthogonal Procrustes projection to align local text and image embeddings, and a blending loss that anchors outputs to natural image statistics. Without altering CLIP's weights, this framework unlocks capabilities such as text-to-image generation, style transfer, and image reconstruction. These findings suggest that discriminative models may hold untapped generative potential, hidden in plain sight.

2025

Understanding Adversarial Training with Energy-based Models*arxiv*

We aim at using Energy-based Model (EBM) framework to better understand adversarial training (AT) in classifiers, and additionally to analyze the intrinsic generative capabilities of robust classifiers. By viewing standard classifiers through an energy lens, we begin by analyzing how the energies of adversarial examples, generated by various attacks, differ from those of the natural samples. The central focus of our work is to understand the critical phenomena of Catastrophic Overfitting (CO) and Robust Overfitting (RO) in AT from an energy perspective. We analyze the impact of existing AT approaches on the energy of samples during training and observe that the behavior of the "delta energy" -- change in energy between original sample and its adversarial counterpart -- diverges significantly when CO or RO occurs. After a thorough analysis of these energy dynamics and their relationship with overfitting, we propose a novel regularizer, the Delta Energy Regularizer (DER), designed to smoothen the energy landscape during training. We demonstrate that DER is effective in mitigating both CO and RO across multiple benchmarks. We further show that robust classifiers, when being used as generative models, have limits in handling trade-off between image quality and variability. We propose an improved technique based on a local class-wise principal component analysis (PCA) and energy-based guidance for better class-specific initialization and adaptive stopping, enhancing sample diversity and generation quality. Considering that we do not explicitly train for generative modeling, we achieve a competitive Inception Score (IS) and Fréchet inception distance (FID) compared to hybrid discriminative-generative models.

2024

Shedding More Light on Robust Classifiers under the lens of Energy-based Models*ECCV2024*

By reinterpreting a robust discriminative classifier as Energy-based Model (EBM), we offer a new take on the dynamics of adversarial training (AT). Our analysis of the energy landscape during AT reveals that untargeted attacks generate adversarial images much more in-distribution (lower energy) than the original data from the point of view of the model. Conversely, we observe the opposite for targeted attacks. On the ground of our thorough analysis, we present new theoretical and practical results that show how interpreting AT energy dynamics unlocks a better understanding: (1) AT dynamic is governed by three phases and robust overfitting occurs in the third phase with a drastic divergence between natural and adversarial energies (2) by rewriting the loss of TRadeoff-inspired Adversarial DEfense via Surrogate-loss minimization (TRADES) in terms of energies, we show that TRADES implicitly alleviates overfitting by means of aligning the natural energy with the adversarial one (3) we empirically show that all recent state-of-the-art robust classifiers are smoothing the energy landscape and we reconcile a variety of studies about understanding AT and weighting the loss function under the umbrella of EBMs. Motivated by rigorous evidence, we propose Weighted Energy Adversarial Training (WEAT), a novel sample weighting scheme that yields robust accuracy matching the state-of-the-art on multiple benchmarks such as CIFAR-10 and SVHN and going beyond in CIFAR-100 and Tiny-ImageNet. We further show that robust classifiers vary in the intensity and quality of their generative capabilities, and offer a simple method to push this capability, reaching a remarkable Inception Score (IS) and FID using a robust classifier without training for generative modeling.

 Courses

2025 Aug – 2025 Aug
Saarbrücken, Germany

CISPA - ELLIS - Summer School 2025

This year's Summer School at CISPA- Helmholtz Center for Information Security will be supported by ELLIS and ELISA

2024 Jun – 2024 Jun
Eindhoven, Netherlands

Generative Modeling Summer School

Eindhoven University of Technology

The summer school is targeted toward PhD students working with data science and/or AI broadly, and for whom generative modeling potentially plays a part in their projects.